FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 17-11-2019

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

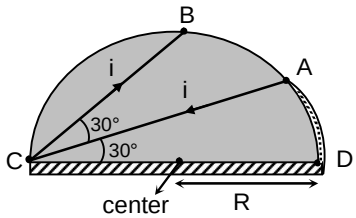
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A current carrying wire is placed in the grooves of an insulating semicircular disc of radius R as shown in the figure. The current enters at point A and leaves from point B. The magnetic field at point D is



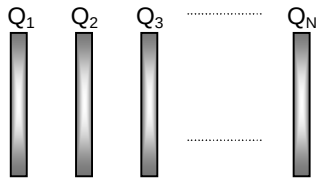
(A) $\frac{\mu_0 i}{4\sqrt{3}\pi R}$

(B) $\frac{\mu_0 i}{8\sqrt{3}\pi R}$

(C) $\frac{\mu_0 i}{2\sqrt{3}\pi R}$

(D) Zero

2. N large square metallic plates are separated by a small distance and arranged parallel to its nearby plates as shown in the figure. Each of the plates are imparted charges Q_1, Q_2 …….. Q_N respectively.
Choose the correct answer(s) from the options.



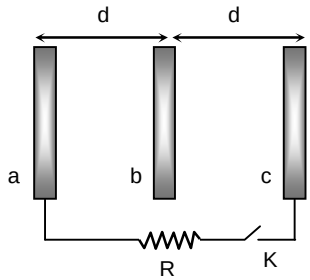
(A) Charges on the outermost surfaces of the plate system are equal and opposite.

(B) If the plate 1 is earthed, then charge flown to the earth is Q_1.

(C) If the plate 3 is earthed, then charge flown to the earth is $\frac{\sum_{i=1}^{N} Q_i}{N}$

(D) If the plate 4 is earthed, the charge flown to the earth is $\sum_{i=1}^{N} Q_i$.

3. Three identical large metal plates each of area 'A' are placed parallel to each other at equal distance d as shown. Initially metal plate 'a' is uncharged, while metal plates b and c have respective charges $+q_0$ and $-q_0$ initially as shown. Metal plates a and c are connected by switch K through resistor of resistance R. The key K is closed at time t = 0. Then the magnitude of current through the resistor at any time t is



(A) $\frac{q_0 d}{RA\varepsilon_0} e^{\frac{-dt}{RA\varepsilon_0}}$

(B) $\frac{q_0 d}{RA\varepsilon_0} e^{\frac{-2dt}{RA\varepsilon_0}}$

(C) $\frac{q_0 d}{2RA\varepsilon_0} e^{\frac{-dt}{RA\varepsilon_0}}$

(D) $\frac{q_0 d}{2RA\varepsilon_0} e^{\frac{-2dt}{RA\varepsilon_0}}$

4. A spherical black body has a radius R and steady surface temperature T, heat sources ensure the heat evolution at a constant rate and distributed uniformly over its volume. What should be the new steady state surface temperature of the object if the radius is decreased by half? Assume surrounding to be at absolute zero and heat evolution rate through unit volume remain same.

(A) $(2)^{\frac{1}{4}}T$

(B) $\sqrt{2}T$

(C) $\dfrac{T}{(2)^{\frac{1}{4}}}$

(D) $\dfrac{T}{\sqrt{2}}$

5. Three charged conducting metal spheres of radius R_1, R_2 and R_3 are connected together by very thin wires and they are separated by large distances from each other. If $R_1 < R_2 < R_3$, at equilibrium, which of the following sets of relations involving the electric field strength E generated by a sphere at a point just outside the surface, its potential V and its charge Q must hold between the spheres.

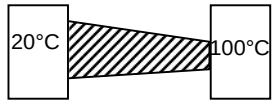
(A) $V_1 = V_2 = V_3$, $E_1 < E_2 < E_3$, $Q_1 < Q_2 < Q_3$

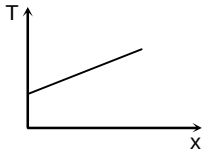
(B) $V_1 = V_2 = V_3$, $E_1 > E_2 > E_3$, $Q_1 < Q_2 < Q_3$

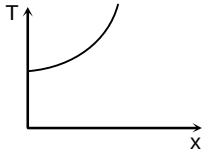
(C) $V_1 < V_2 < V_3$, $E_1 < E_2 < E_3$, $Q_1 = Q_2 = Q_3$

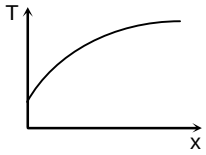
(D) $V_1 > V_2 > V_3$, $E_1 < E_2 < E_3$, $Q_1 > Q_2 > Q_3$

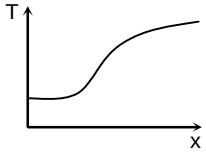
6. A conical metal beam is placed between two constant temperature surface. The surfaces are at 20°C and 100°C as shown in the figure. Which one of the following graphs best represents the temperature along the axis of the beam at a distance 'x' from the left end in steady state?



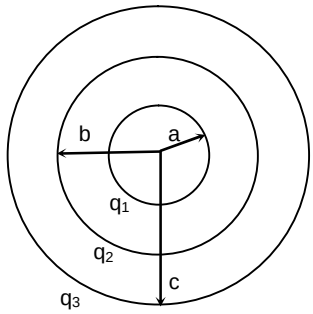
(A)


(B)


(C)


(D)


7. Three concentric thin shells are shown carrying charges q_1, q_2 and q_3 and of radii a, b and c respectively. If the middle shell expands from radius b to b′ (b′ < c). The work done by the electric field in the process is



(A) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q_2^2}{2b'}-\frac{q_2^2}{2b}+\frac{q_1q_2}{b'}-\frac{q_1q_2}{b}\right)$

(B) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q_2^2}{2b}-\frac{q_2^2}{2b'}+\frac{q_1q_2}{b}-\frac{q_1q_2}{b'}\right)$

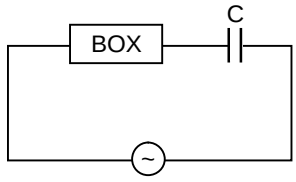
(C) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q_2^2}{2b'}-\frac{q_2^2}{2b}\right)$

(D) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q_1q_2}{b}-\frac{q_1q_2}{b'}\right)$

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

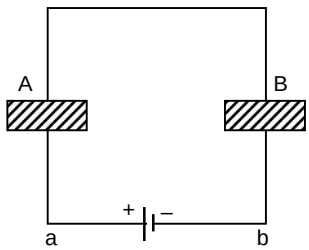
8. A box has a large electric circuit inside it. When it was connected to an AC generator, it was found that it was putting a lot of load on the generator and the power factor of the box was $\frac{1}{\sqrt{2}}$. Now a capacitor of capacitance C was connected in series with the box and the power factor of the circuit becomes equal to 1. Then



AC generator of angular frequency ω

(A) The impedance of the box must be resistive and capacitive.
(B) The impedance of the box must be resistive and inductive.
(C) The impedance of the box is $\frac{\sqrt{2}}{\omega C}$
(D) The impedance of the box is $\frac{1}{\sqrt{2}\,\omega C}$

9. Consider two identical dielectric slabs that are inserted into two identical capacitors A and B connected to the battery as shown in the figure. Now the dielectric slab of capacitor B is pulled out slowly from it with the battery still remaining connected. Then during the process of removing the dielectric slab, we observe the following statement(s) to be correct.



(A) Positive charge flows from a to b.
(B) Final charge on capacitor B will be less than that on capacitor A.
(C) Work is done by the external force F appears completely as heat in the circuit.
(D) Internal energy of the battery increases.

10. The relation between internal energy U, pressure P and Volume V of a gas in an adiabatic process U = a + b PV, where a and b are positive constants. Then

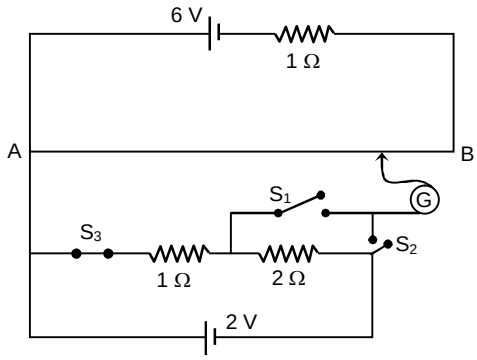
(A) Equation for the process is $PV^{\frac{b+2}{b}}$ = constant

(B) Equation for the process is $PV^{\frac{b+1}{b}}$ = constant

(C) The gas must be monoatomic

(D) The gas must be diatomic

11. Figure shows a potentiometer circuit. The length of potentiometer wire AB is 120 cm and its resistance is 2 Ω. The internal resistance of 2V cell is negligible and the internal resistance of 6V cell is 1 Ω initially the switches S_1 and S_2 are open and S_3 is closed.



(A) When only S_2 is open, balanced length $\ell = 20\,\text{cm}$

(B) When only S_1 is open, balanced length $\ell = 60\,\text{cm}$

(C) When all three switches are closed, balanced length is 60 cm

(D) When S_3 is open, S_2 is also open but S_1 is closed, balance length is 60 cm

12. Two identical objects A and B are at temperatures T_A and T_B respectively. Both objects are placed in a room with perfectly absorbing walls maintained at a temperature. T($T_A > T > T_B$). The objects A and B attain the temperature T eventually. Select the correct statements from the following

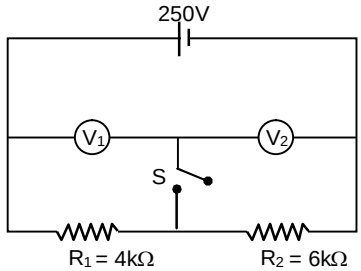
(A) A only emits radiation, while B only absorbs it until both attain the final temperature T.

(B) A loses more heat by radiation than it absorbs, while B absorbs more radiation than it emits, until they attain the temperature T.

(C) Both A and B only absorbs radiation, but do not emit it, until they attain the temperature T.

(D) Each object continues to emit and absorb radiation even after attaining the temperature T.

13. In the circuit shown in figure, V_1 and V_2 are two voltmeters having resistances 6 KΩ and 4 KΩ respectively. The emf of the battery is 250 V, having negligible internal resistance. The resistances R_1 = 4 KΩ and R_2 = 6 KΩ are also connected in the circuit as shown. Choose the correct option(s).



(A) When switch S is open, then readings of voltmeters are V_1 = 100 V and V_2 = 150 V.

(B) When switch S is open, then readings of voltmeters are V_1 = 150 V and V_2 = 100 V.

(C) When switch S is closed, then readings of voltmeter are V_1 = 100 V and V_2 = 150 V.

(D) When switch S is closed, then readings of voltmeters are V_1 = 125 V and V_2 = 125 V.

14. A charged particle 'P' leaves the origin with speed $v = v_0$ at some inclination with the x-axis. There is a uniform magnetic field B along the x-axis. The particle 'P' strikes a fixed target 'T' on the x-axis for a minimum value of $B = B_0$. The particle 'P' will also strike the target T, if
(A) $B = 2B_0, v = 2v_0$
(B) $B = 2B_0, v = v_0$
(C) $B = B_0, v = 2v_0$
(D) $B = \frac{B_0}{2}, v = 2v_0$

Paragraph Type (One Option Correct)

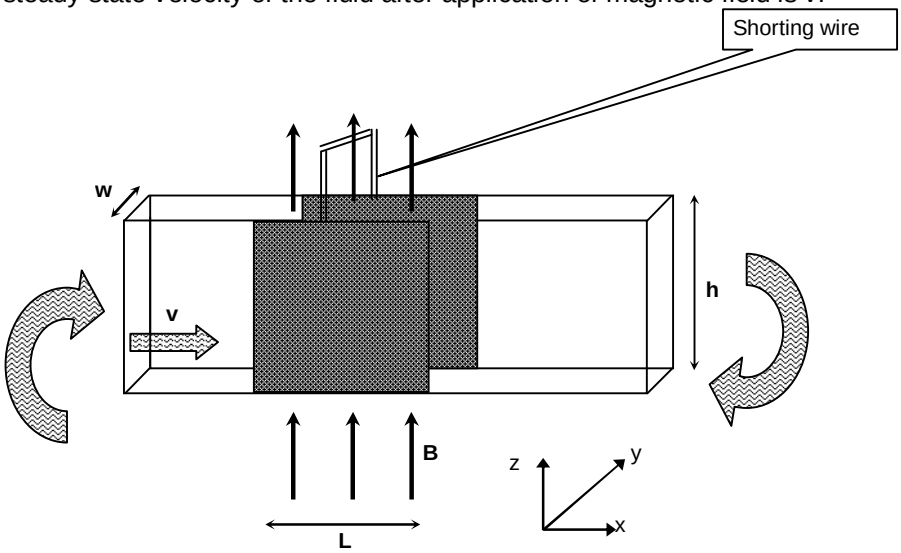
This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

The production of electrical power through the use of a conducting fluid moving through a magnetic field is referred to as a MAGNETO HYDRODYNAMIC (MHD) GENERATOR. A simplistic way of functioning of MHD is shown in the figure. In the figure shown, a horizontal rectangular plastic pipe of width w and height h is filled with mercury of electrical conductivity σ. A pressure difference ΔP is produced by a turbine which drives this fluid with a constant speed V_0. The two opposite vertical walls of a section of the pipe with length L are made of copper.
To simplify the situation we assume the following.

- Although the fluid is viscous, its speed is uniform over the entire cross-section.
- The speed of the fluid is always proportional to the net external force acting upon it.
- The fluid is incompressible.

These conducting walls are electrically shorted with an external wire as shown in the figure. The steady state velocity of the fluid after application of magnetic field is v.



15. The current flowing in the shorting wire in the steady state is
(A) $BLh\sigma v$
(B) $BLw\sigma v$
(C) $\frac{BL^2h\sigma v}{w}$
(D) $Bvh\sigma w$

16. Taking the assumption taken in the MHD working that speed of the fluid is always proportional to the net external force acting upon it, the expression of steady state velocity v after application of magnetic field is

(A) $\dfrac{\Delta P}{B^2 L\sigma}$

(B) $\dfrac{\Delta P v_0}{\Delta P + B^2 L\sigma v_0}$

(C) $\dfrac{\Delta P h v_0}{\Delta P h + B^2 L w\sigma v_0}$

(D) $\dfrac{\Delta P v_0}{\Delta P + B^2 w\sigma v_0}$

Paragraph for Question Nos. 17 and 18

A nonlinear circuit element can be made out of a parallel plate capacitor and small balls, each of mass m, that can move between the plates. The balls collide inelastically with the plates, dissipate all kinetic energy as thermal energy and immediately release all the charge they are carrying to the plate. Almost instantaneously, the balls then pick up a small charge of magnitude q from the plate, the balls are then repelled directly towards the other plate under electrostatic forces only. Another collision happens, kinetic energy is dissipated, the balls give up the charge, collect a new charge and the cycle repeats. There are n_0 balls per unit surface area of the plates. The capacitor has a capacitance C, the separation 'd' between the plates is much larger than the radius r of the balls. A battery is connected to the plates in order to maintain a constant potential difference V. Neglect edge effects and assume magnetic and gravitational forces to be negligible. (ε_0 is permittivity of free space)

17. The current between the plates in terms of any or all of the given parameters ε_0, m, n_0, c, q and V is

(A) $\dfrac{2Cn_0 q}{\varepsilon_0}\sqrt{\dfrac{qV}{m}}$

(B) $\dfrac{Cn_0 q}{\varepsilon_0}\sqrt{\dfrac{2qV}{m}}$

(C) $\dfrac{Cn_0 q}{\varepsilon_0}\sqrt{\dfrac{qV}{m}}$

(D) $\dfrac{Cn_0 q}{\varepsilon_0}\sqrt{\dfrac{qV}{2m}}$

18. The effective resistance of the device in terms of any or all of the given parameters ε_0, m, n_0, c, q and V is

(A) $\dfrac{\varepsilon_0}{Cn_0 q}\sqrt{\dfrac{mV}{q}}$

(B) $\dfrac{\varepsilon_0}{Cn_0 q}\sqrt{\dfrac{2mV}{q}}$

(C) $\dfrac{\varepsilon_0}{Cn_0 q}\sqrt{\dfrac{mV}{2q}}$

(D) $\dfrac{2\varepsilon_0}{Cn_0 q}\sqrt{\dfrac{mV}{q}}$

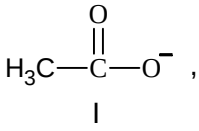
Chemistry

PART – II

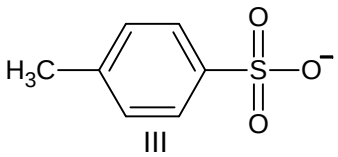
SECTION – A
(One Options Correct Type)

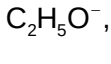
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

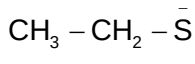
19. The order of increasing nucleophilicity of the anions (in polar protic solvent)


I


II

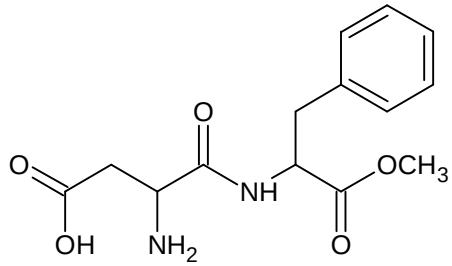

III


IV


V

(A) V < I < III < II < IV
(B) III < I < IV < II < V
(C) III < II < I < IV < V
(D) II < III < I < V < IV

20. Aspartame is an artificial non-saccharide sweetner 200 times sweeter than sucrose and is commonly used as a sugar substitute in foods and beverages. Structure of Aspartame is given below:

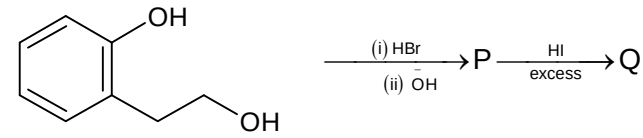


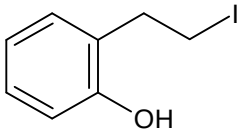
Which of the following is correct about aspartame?
(A) It contains 4 functional groups and on hydrolysis produce 2 essential amino acids
(B) It contains 5 functional groups and on hydrolysis produce one essential and one non essential amino acids
(C) It contains 4 functional group and on hydrolysis produce 2 non essential amino acids.
(D) It contains a 4 functional groups and on hydrolysis produce 1 essential and 1 non essential amino acids.

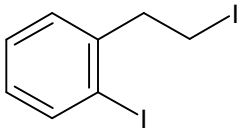
21. When trans-2 methyl cyclopentanol is treated with tosyl chloride and the product with potassium tert-butoxide; then the final product would be
(A) Methyl-cyclopentene
(B) 3-methyl cyclopentene
(C) Mixture of A and B
(D) t-butoxy cyclopentane

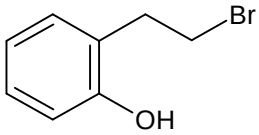
22. Correct increasing order of acidic strength among
Formic acid (I), Acetic acid (II), Oxalic acid (III) and Malonic acid (IV) would be:
(A) II < I < IV < III
(B) II < IV < I < III
(C) I < II < IV < III
(D) III < IV < II < I

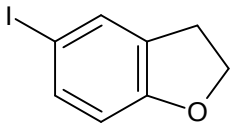
23. Upon treatment with sulphuric acid, a mixture of tert-butyl alcohol and ethyl alcohol at room temperature gives mainly
(A) diethyl ether
(B) iso butane
(C) Di-t-butyl ether
(D) Tert-butyl ethyl ether

24. 

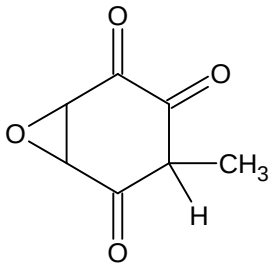
(A) 

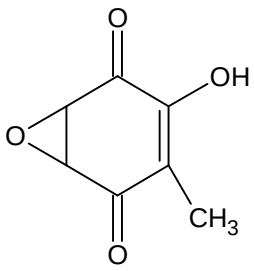
(B) 

(C) 

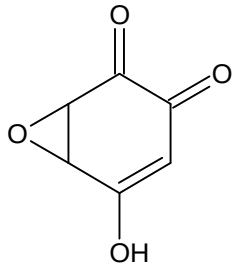
(D) 

25. Terreic acids ($C_7H_6O_4$) is a naturally occurring antibiotic substance. Which of the following structure is most stable form of terreic acid

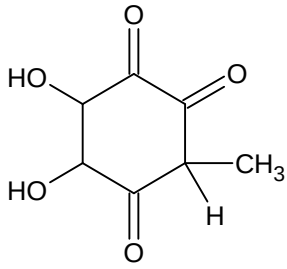
(A) 

(B) 

(C)



(D)

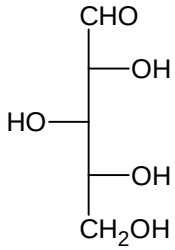


(One or More than one correct type)

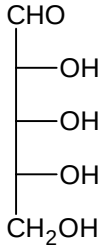
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. Which of the following compound/s is/are a D-sugar that gives an optically active pentitol on reduction?

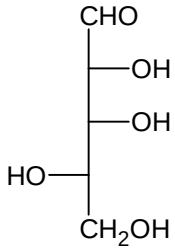
(A)



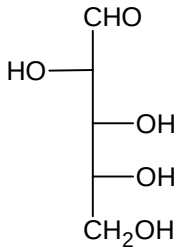
(B)



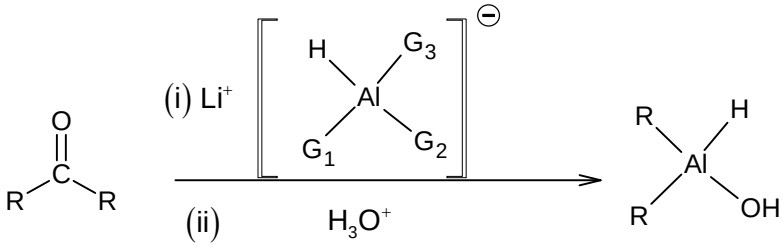
(C)



(D)



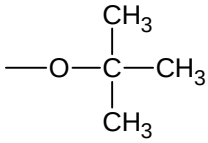
27.



Which of the following statement(s) is/are correct about the above reaction

(A) Rate of reaction decreases as the size of R increases

(B) Reaction rate is more when $G_1 = G_2 = G_3 = H$ (LAH) than When $G_1 = G_2 = G_3 =$

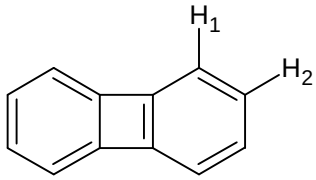


(LTBA)

(C) LTBA is much more sterically hindered compared to LAH and therefore, have difficulty in transferring hydride ions.

(D) LTBA is used to selectively reduce aldehydes and ketones in the presence of esters.

28.



Biphenylene

Which of the following statement(s) is/are correct about Biphenylene?

(A) It is a dimer of reactive intermediate benzyne

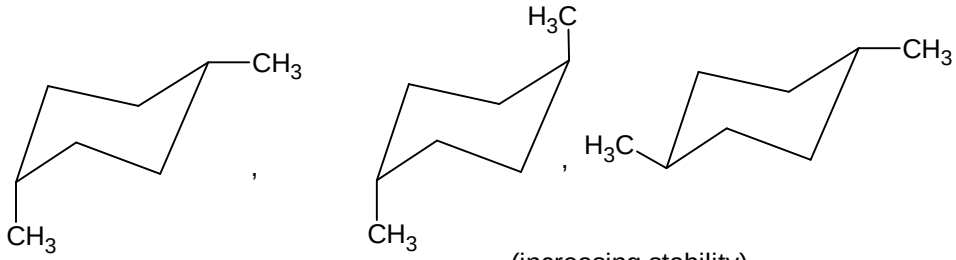
(B) All C – C bonds have equal length

(C) H_1 is more acidic than H_2

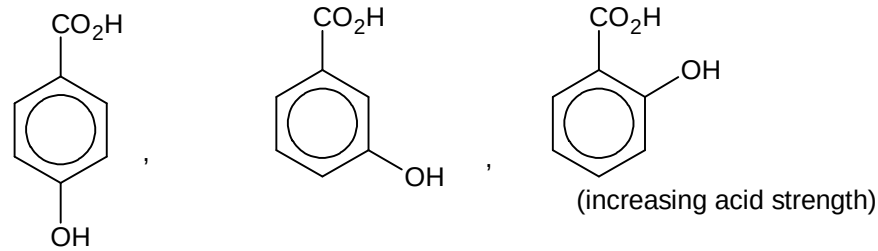
(D) Due to the influence of central ring it has lesser degree of aromaticity.

29. Which of the following is/are arranged in correct mentioned order

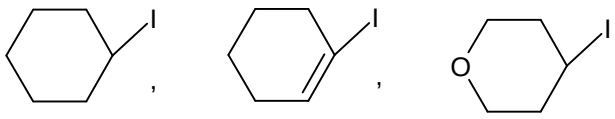
(A)



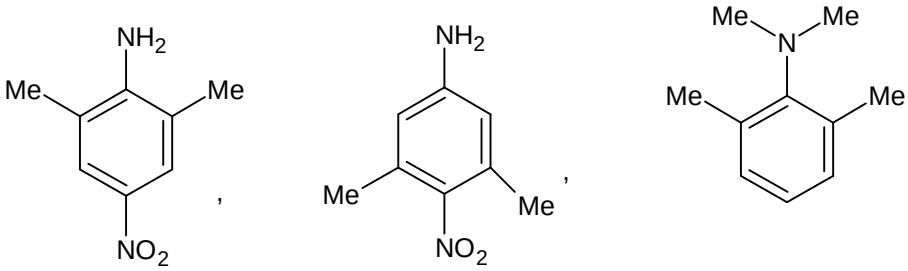
(increasing stability)

(B) 

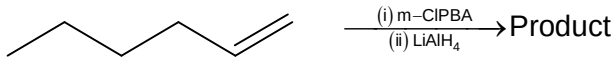
(increasing acid strength)

(C) 

(increasing rate of solvolysis)

(D) 

(increasing basic strength)

30. 

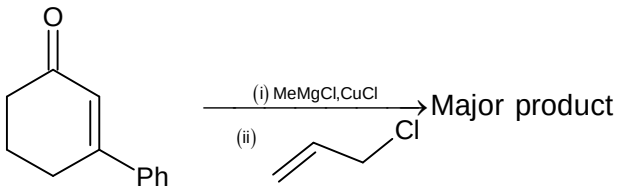
Which of the following is/are correct about product?

(A) It is racemic mixture

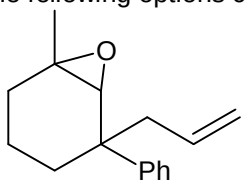
(B) Product is a hydrocarbon which can not decolourise Br_2 – water

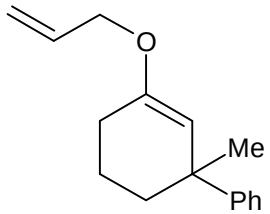
(C) Product can give positive iodoform test

(D) Product is also obtained from the alkene by HBO (hydroboration – oxidation) process

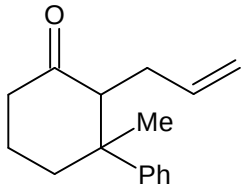
31. 

Which of the following options can not be the major product?

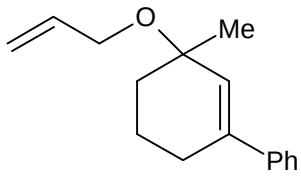
(A) 

(B) 

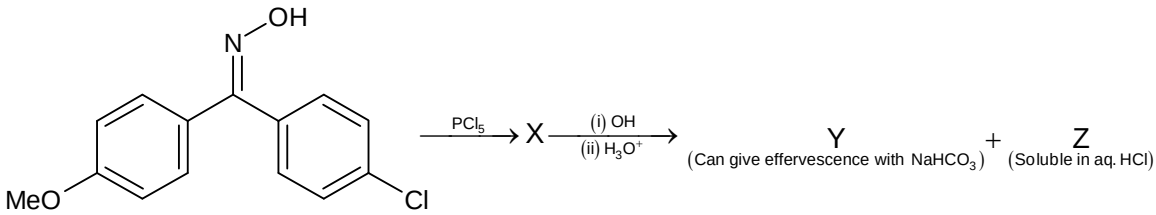
(C)



(D)



32.



Identity the correct statements:

(A) (Y) is more acidic than benzoic acid

(B) (Z) is weaker base than aniline

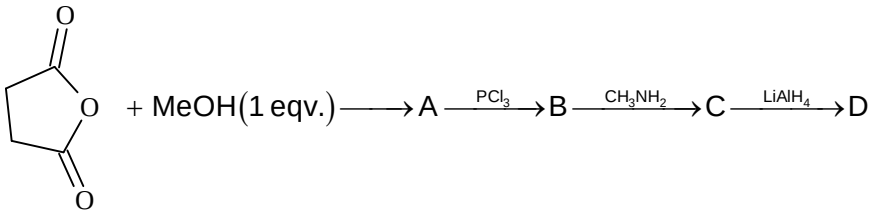
(C) In (X) both the benzene rings are activated towards EAS

(D) Z reacts at much faster rate towards electrophile than Y.

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 33 and 34



33. What would be correct choice for 'D' from the following statements:

(A) On heating with $CHCl_3$ along with aq. KOH produce foul smelling compound.

(B) Produce blue colour in Victor Meyor test.

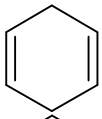
(C) On treatment with HNO_2 produces yellow oily compound.

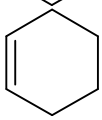
(D) It can produce substituted ammonium nitrite salt on treatment with HNO_2.

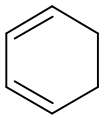
34. $D \xrightarrow[\text{(ii) AgOH,}\Delta]{\text{(i) } CH_3I,\ \text{excess}} E \xrightarrow[\text{(ii) AgOH,}\Delta]{\text{(i) } CH_3I,\ \text{excess}} F + G(C_3H_9N)$

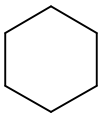
$F \xrightarrow{CH_2=CH_2,\ \Delta} H$

Structure of H would be:

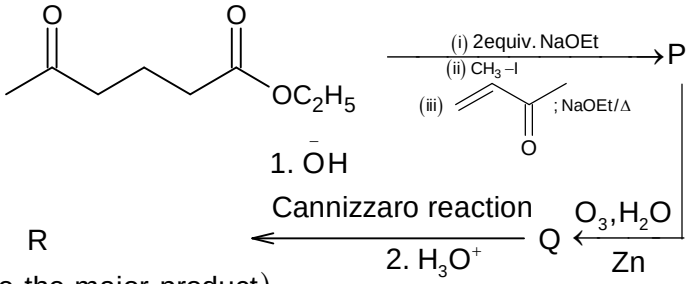
(A) 

(B) 

(C) 

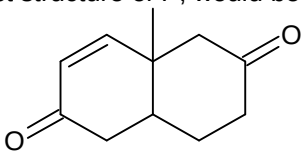
(D) 

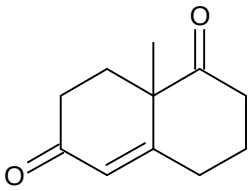
Paragraph for Question Nos. 35 to 36

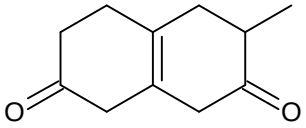


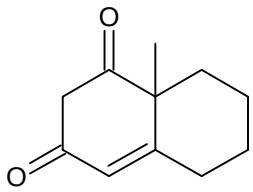
(may not be the major product)

35. The correct structure of P, would be

(A) 

(B) 

(C) 

(D) 

36. INCORRECT statement about R is:
(A) It produces effervescence on addition of $NaHCO_3$.
(B) It give positive iodoform test.
(C) It give coloured compound with 2, 4 DNPH.
(D) It gives reddish brown colouration with freshly prepared $FeCl_3$ solution.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Let $f(x) = \left[2019^{\sin^2 x} + 2019^{\cos^2 x}\right]$ (where [.] denotes the greatest integer function), then number of integers in range of f(x) is
(A) 1930
(B) 1931
(C) 1932
(D) none of these

38. Circle $x^2 + y^2 - 2x - 3 = 0$ intersects parabola $y^2 = 4x$ at points A and B. Let tangents to the parabola at A and B intersect at point C. If radius of circle increases at a rate of 1 m/s, the speed of point C after 2019 seconds is (in m/s)
(A) 1
(B) $\frac{2019}{2}$
(C) 2019
(D) none of these

39. Let S_1 and S_2 be two externally tangent circles having the same radius 'r'. Let L be a direct common tangent to S_1 and S_2. If S_n be circle such that it touches S_{n-1} and S_1 externally and line L, $\forall\ n \geq 3$, $n \in N$, then radius of S_{2020} is
(A) r
(B) $\frac{r}{2019}$
(C) $\frac{r}{(2019)^2}$
(D) $\frac{r}{(2019)^3}$

40. In a scalene ΔABC, E and F are points on the sides AC and AB respectively such that BE and CF are internal angle bisectors of the triangle. If BE and CF intersect at I and IE = IF, then $\angle CAB$ can be equal to
(A) $\frac{\pi}{3}$
(B) $\frac{\pi}{2}$
(C) $\frac{2\pi}{3}$
(D) $\frac{3\pi}{4}$

41. Let A = {x | x ∈ N, x ≤ 105, x is coprime with 105} value of $\sum_{r\in A}\cos\left(\frac{2\pi r}{105}\right)$ is equal to

(A) –3
(B) –1
(C) 0
(D) none of these

42. The trigonometric equation $\sum_{r=1}^{6} r^3 \sin(r\pi x) = 441$ ($x \in R$) has

(A) no solution
(B) unique solution
(C) infinite solutions
(D) finite number of solutions

43. Value of $\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+2\cos 1360^\circ}}}}$ is equal to

(A) 2 cos 5°
(B) 2 cos 25°
(C) 2 cos 65°
(D) 2 cos 85°

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

44. Let $f(x) = \sum_{r=0}^{n}(r+2)x^{2r}$ and $g(x) = ax^2 + \cos^2 b + \sin b$ where $n \in N$; $a, b \in R$. If d(n, a, b) represents shortest distance between curves y = f(x) and y = g(x) for given values of n, a, b ; then

(A) d(2019, 2020, b) = $\frac{3}{4}$ $\forall$ b ∈ R

(B) d(n, –2018, b) = $\frac{3}{4}$ $\forall$ b ∈ R, n ∈ N

(C) d(n, a, b) = $\frac{3}{4}$ $\forall$ n ∈ N, a, b ∈ R

(D) d(2020, a, b) = $\frac{3}{4}$ $\forall$ a, b ∈ R

45. For the hyperbola $2x^2 + 12xy - 7y^2 - 16x + 2y - 3 = 0$, which of the following is/are true?

(A) Equation of its conjugate hyperbola is $2x^2 + 12xy - 7y^2 - 16x + 2y + 17 = 0$
(B) Its asymptotes are given by $2x^2 + 12xy - 7y^2 - 16x + 2y + 7 = 0$
(C) Its principal axes are given by $2x^2 - 2y^2 - x + 7y - 3xy - 3 = 0$
(D) Orthocentre of triangle formed by any three points on the hyperbola lies on the hyperbola itself

46. Let P be a point on the parabola $y^2 = 4x$ such that normal chord drawn at P to parabola has shortest length. If tangent and normal at P intersects axis of parabola at A and B respectively, then

(A) Slope of normal at P is ±2
(B) Slope of tangent at P is $\pm\frac{1}{\sqrt{2}}$
(C) Area of ΔPAB is $12\sqrt{2}$ sq. units
(D) Circumcenter of ΔPAB is (1, 0)

47. A particle starts from origin and moves at a constant speed of 4 unit/s towards positive x-axis. The particle changes its direction by angle 'θ' (clockwise or anticlockwise at random) with respect to its previous direction after every 2 seconds, then

(A) For $\theta = \frac{\pi}{2}$, particle can pass through the point (2019, 0)

(B) For $\theta = \pi$, particle will trace a line segment

(C) For $\theta = \frac{\pi}{3}$, particle may lie on the line $y + \sqrt{3}x = 0$

(D) For $\theta = \frac{\pi}{6}$, after 2019 seconds, particle may lie on the line $x - \sqrt{3}y - 8 = 0$

48. Let $A = \{x \mid 1 + 4^{\sin x - \cos x} = \sec^2 x, x > 0\}$, then

(A) elements of set A form an Arithmetic Progression

(B) number of elements of set A which are less than $n\pi$ is $n + 1$

(C) the sum of smallest four elements is 3π

(D) set A contains only irrational elements

49. A square is formed such that its opposite sides are tangential to the hyperbola $x^2 - y^2 = 1$. Determine which of the following is/are true?

(A) Side length of largest possible square is 2

(B) Side length of largest possible square is $2\sqrt{2}$

(C) Side length of smallest possible square is $2^{\frac{3}{4}}$

(D) Side length of smallest possible square is $2^{\frac{5}{4}}$

50. For given pair of lines $f(x, y) = ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$, which of the following is/are **NOT** true?

(A) If $h^2 > ab$, then $f(x, y) = 0$ always represents distinct real lines

(B) If $h^2 = ab$, then $f(x, y) = 0$ always represents coincident real lines

(C) If $h^2 = ab$, then $f(x, y) = 0$ always represents distinct real lines

(D) If $h^2 \geq ab$, then $f(x, y) = 0$ always represents real lines

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 to 52

Read the following write up carefully and answer the following questions:

Consider the lines $x = 0$, $y = 0$ and $3x + 4y = 12$. Let S_1, S_2, S_3, S_4 be four distinct circles with each of them touching the three lines. Let S_1 be the smallest of the four circles. Let A_i and C_i represent the area and the centre respectively of circle S_i for $i = 1, 2, 3, 4$.

51. Which of the following is true for A_1, A_2, A_3, A_4?

(A) Their Arithmetic mean is 25π

(B) Their Geometric mean is 6π

(C) Their Harmonic mean is 2π

(D) None of these

52. Let circles S_5, S_6, S_7 be drawn with C_2C_3, C_3C_4 and C_4C_2 as diameters, then radical centre of the circles S_5, S_6, S_7 is
(A) (1, 1)
(B) (0, 0)
(C) $\left(2, \frac{3}{2}\right)$
(D) $\left(\frac{4}{3}, 1\right)$

Paragraph for Question Nos. 53 to 54

Read the following write up carefully and answer the following questions:
Let ABC be a triangle such that $\angle ABC = 75°$, $\angle BCA = 60°$, $\angle CAB = 45°$ and AB = 1 unit. Let D, E, F be points on sides AB, BC, AC respectively such that perimeter of ΔDEF is minimum

53. Perimeter of ΔDEF is
(A) $\frac{3}{2}$
(B) $\frac{\sqrt{3}-1}{2}$
(C) $\frac{\sqrt{3}+1}{2}$
(D) none of these

54. Radius of circumcircle of ΔDEF is equal to
(A) $\sqrt{3}-1$
(B) $\frac{1}{2\sqrt{3}}$
(C) $\frac{1}{\sqrt{3}}$
(D) none of these